



Cullen College of Engineering
UNIVERSITY OF HOUSTON

Sampling & Data

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Statistics

- The science of collecting, organizing, analyzing, and interpreting data in order to make decisions.

Population

- In statistics, we generally want to study a population. You can think of a population as a collection of persons, things, or objects under study.

Population

- Determine the population: We want to know the average (mean) amount of money first year college students spend at the University of Houston Engineering Academy on school supplies that do not include books. We randomly survey 100 first year students at the college. Three of those students spent \$150, \$200, and \$225, respectively.

Population

- The population is all first-year students attending HCC/UH this term.

Sample

- To study the population, we select a sample. The idea of sampling is to select a portion (or subset) of the larger population and study that portion (the sample) to gain information about the population.

Sample

- Determine the sample: We want to know the average (mean) amount of money first year college students spend at University of Houston on school supplies that do not include books. We randomly survey 100 first year students at the college. Three of those students spent \$150, \$200, and \$225, respectively.
- The sample could be all students enrolled in one section of a beginning statistics course.

Sample

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Parameter

- A parameter is a number that is a property of the population. Because it takes a lot of time and money to examine an entire population, sampling is a very practical technique.

Parameter

- Determine the parameter: We want to know the average (mean) amount of money first year college students spend at University of Houston on school supplies that do not include books. We randomly survey 100 first year students at the college. Three of those students spent \$150, \$200, and \$225, respectively.

Parameter

- The parameter is the average (mean) amount of money spent (excluding books) by first year college students at University of Houston this term.

Statistic

- A statistic is a number that represents a property of the sample. For example, if we consider one math class to be a sample of the population of all math classes, then the average number of points earned by students in that one math class at the end of the term is an example of a statistic.

Statistic

- Determine the statistic: We want to know the average (mean) amount of money first year college students spend at University of Houston on school supplies that do not include books. We randomly survey 100 first year students at the college. Three of those students spent \$150, \$200, and \$225, respectively.

Statistic

- The statistic is the average (mean) amount of money spent (excluding books) by first year college students in the sample.

Variables

- A variable, notated by capital letters such as X and Y , is a characteristic of interest for each person or thing in a population.

Variables

- Determine the variable: We want to know the average (mean) amount of money first year college students spend at University of Houston on school supplies that do not include books. We randomly survey 100 first year students at the college. Three of those students spent \$150, \$200, and \$225, respectively.

Variables

- The variable could be the amount of money spent (excluding books) by one first year student. Let X = the amount of money spent (excluding books) by one first year student attending University of Houston.

Variables

- Variables may be **numerical** or **categorical**. Numerical variables take on values with equal units such as weight in pounds and time in hours. Categorical variables place the person or thing into a category. If we let X equal the number of points earned by one math student at the end of a term, then X is a numerical variable. If we let Y be a person's party affiliation, then some examples of Y include Republican, Democrat, and Independent. Y is a categorical variable.

Data

- Data are the actual values of the variable. They may be numbers or they may be words. Datum is a single value.

Data

- Determine the data: We want to know the average (mean) amount of money first year college students spend at University of Houston on school supplies that do not include books. We randomly survey 100 first year students at the college. Three of those students spent \$150, \$200, and \$225, respectively.

Data

- The data are the dollar amounts spent by the first-year students. Examples of the data are \$150, \$200, and \$225.

Data Types

- Data may come from a population or from a sample. Small letters like x or y generally are used to represent data values. Most data can be put into the following categories:
 - Qualitative
 - Quantitative

Qualitative data

- Qualitative data are the result of categorizing or describing attributes of a population. Hair color, blood type, ethnic group, the car a person drives, and the street a person lives on are examples of qualitative data. Qualitative data are generally described by words or letters.

Quantitative data

- Quantitative data are always numbers. Quantitative data are the result of counting or measuring attributes of a population. Amount of money, pulse rate, weight, number of people living in your town, and number of students who take statistics are examples of quantitative data. Quantitative data may be either discrete or continuous.

Discrete data

- All data that are the result of counting are called quantitative discrete data. These data take on only certain numerical values. If you count the number of phone calls you receive for each day of the week, you might get values such as zero, one, two, or three.

Continuous data

- All data that are the result of measuring are quantitative continuous data assuming that we can measure accurately. If you and your friends carry backpacks with books in them to school, the numbers of books in the backpacks are discrete data and the weights of the backpacks are continuous data.

Example

- You go to the supermarket and purchase three cans of soup (19 ounces) tomato bisque, 14.1 ounces lentil, and 19 ounces Italian wedding, two packages of nuts (walnuts and peanuts), four different kinds of vegetable (broccoli, cauliflower, spinach, and carrots), and two desserts (16 ounces Cherry Garcia ice cream and two pounds 32 ounces chocolate chip cookies). Name data sets that are quantitative discrete, quantitative continuous, and qualitative.

One Possible Solution:

- The three cans of soup, two packages of nuts, four kinds of vegetables and two desserts are quantitative discrete data because you count them.
- The weights of the soups (19 ounces, 14.1 ounces, 19 ounces) are quantitative continuous data because you measure weights as precisely as possible.
- Types of soups, nuts, vegetables and desserts are qualitative data because they are categorical.

Example

- Determine the correct data type (quantitative or qualitative). Indicate whether quantitative data are continuous or discrete.
 - a. the number of pairs of shoes you own
 - b. the type of car you drive
 - c. the distance it is from your home to the nearest grocery store
 - d. the number of classes you take per school year.
 - e. the type of calculator you use
 - f. weights of sumo wrestlers
 - g. number of correct answers on a quiz
 - h. IQ scores (This may cause some discussion.)

Displaying Data

- Tables are a good way of organizing and displaying data. But graphs can be even more helpful in understanding the data. There are no strict rules concerning which graphs to use. Two graphs that are used to display qualitative data are pie charts and bar graphs.

De Anza College

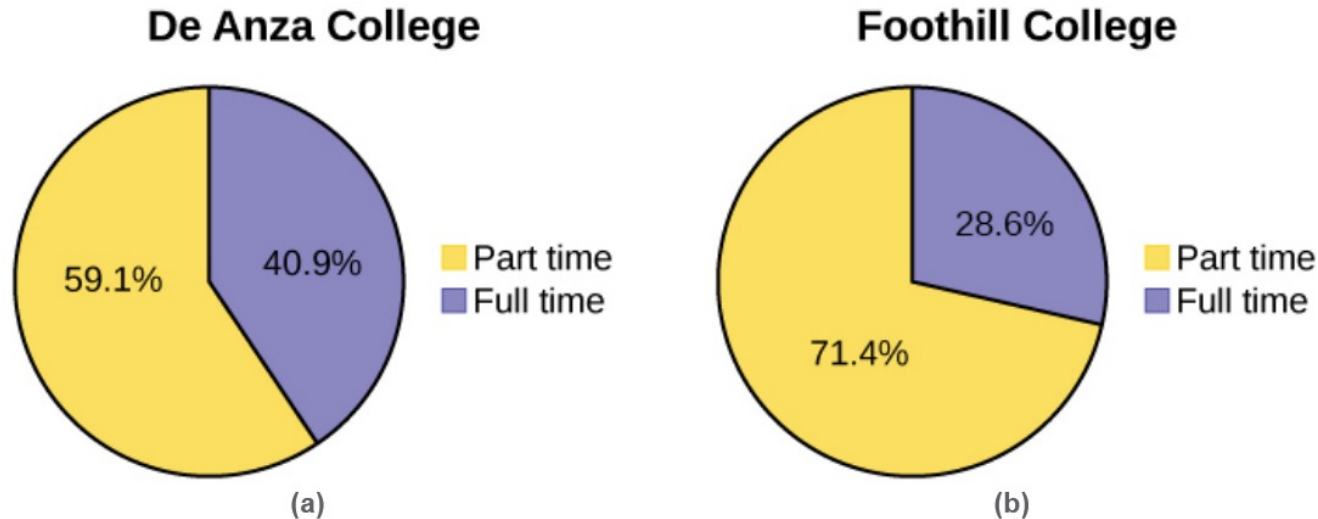
	Number	Percent
Full-time	9,200	40.9%
Part-time	13,296	59.1%
Total	22,496	100%

Foothill College

	Number	Percent
Full-time	4,059	28.6%
Part-time	10,124	71.4%
Total	14,183	100%

Pie Chart

- In a pie chart, categories of data are represented by wedges in a circle and are proportional in size to the percent of individuals in each category.



Student Status

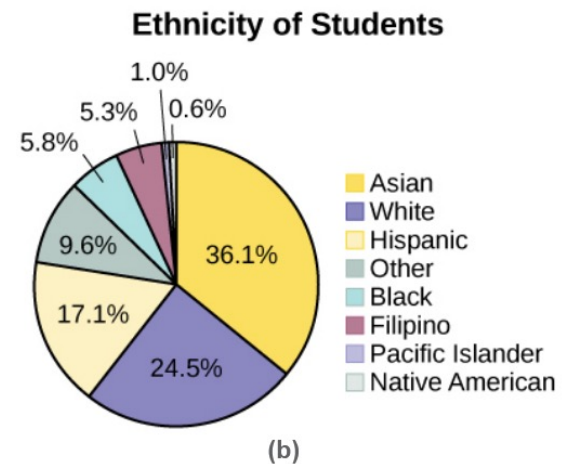
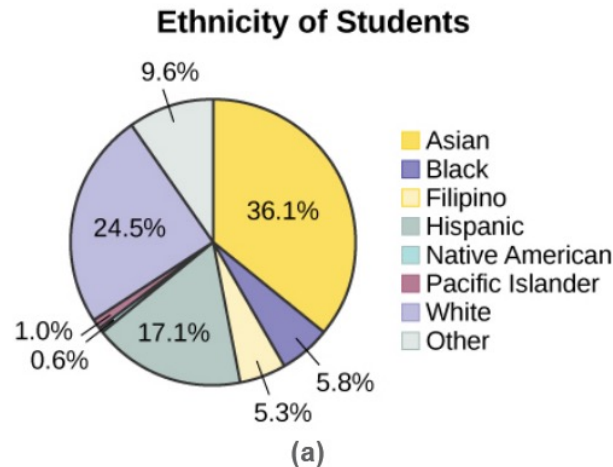
Pie Chart

- Sometimes percentages add up to be more than 100% (or less than 100%). In the graph below, the percentages add to more than 100% because students can be in more than one category.

Characteristic/Category	Percent
Full-Time Students	40.9%
Students who intend to transfer to a 4-year educational institution	48.6%
Students under age 25	61.0%
TOTAL	150.5%

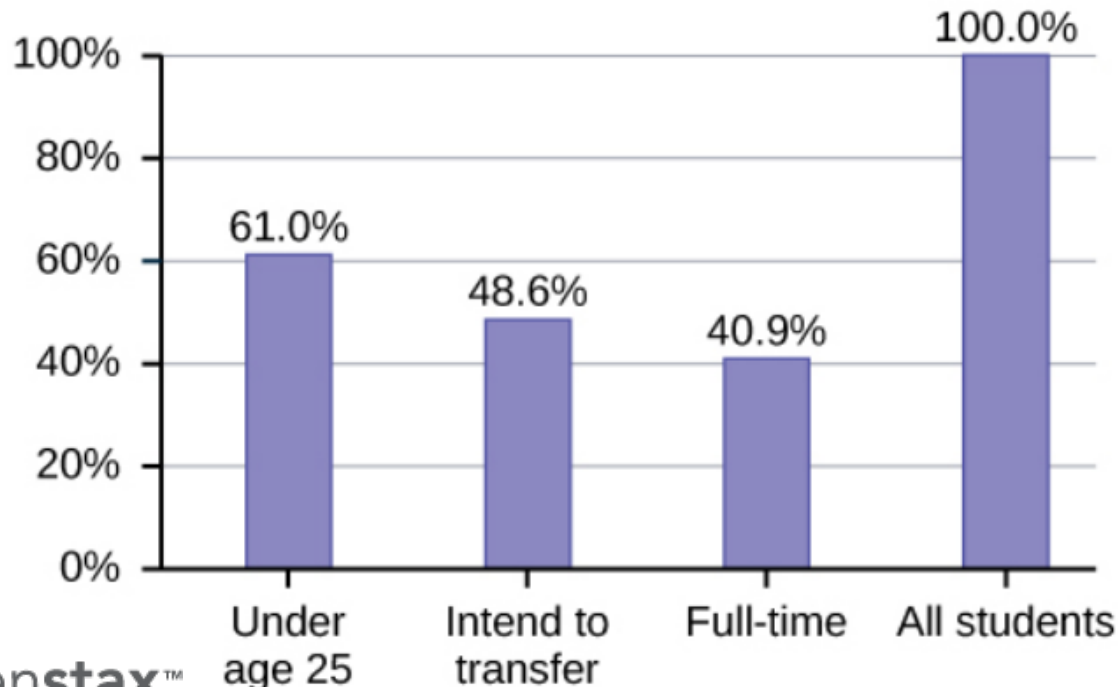
Pie Chart

- The following pie charts have the “Other/Unknown” category included (since the percentages must add to 100%). Chart b is organized by the size of each wedge, which makes it a more visually informative graph than the unsorted, alphabetical graph a.



Bar Graph

- A bar graph is appropriate to compare the relative size of the categories. A pie chart cannot be used. It also could not be used if the percentages added to less than 100%.



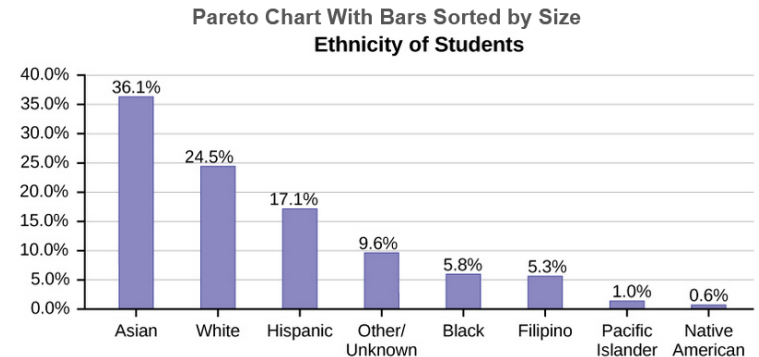
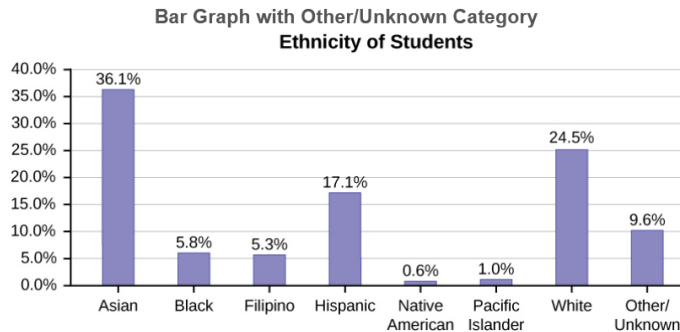
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Bar Graph

- The particular bar graph below can be difficult to understand visually. The other graph is a Pareto chart. The Pareto chart has the bars sorted from largest to smallest and is easier to read and interpret.



Sampling

- Gathering information about an entire population often costs too much or is virtually impossible. Instead, we use a sample of the population. A sample should have the same characteristics as the population it is representing. Most statisticians use various methods of random sampling in an attempt to achieve this goal.

Simple Random Sampling

- A straightforward method for selecting a random sample; give each member of the population a number. Use a random number generator to select a set of labels. These randomly selected labels identify the members of your sample.
- For example, suppose Lisa wants to form a four-person study group from her pre-calculus class, which has 31 members. To choose a simple random sample of size three from the other members of her class, Lisa could put all 30 names in a hat, shake the hat, close her eyes, and pick out three names. A more technological way is for Lisa to first list the last names of the members of her class together with a two-digit number



Stratified Sample

- To choose a stratified sample, divide the population into groups called strata and then take a proportionate number from each stratum.
- For example, you could stratify (group) your college population by department and then choose a proportionate simple random sample from each stratum (each department) to get a stratified random sample. Suppose there are 5 departments, and you want to choose a sample of 50. You choose 10 from each department using SRS sampling.

Cluster Sample

- To choose a cluster sample, divide the population into clusters (groups) and then randomly select some of the clusters. All the members from these clusters are in the cluster sample.
- For example, if you randomly sample four departments from your college population, the four departments make up the cluster sample. Divide your college faculty by department. The departments are the clusters. Number each department, and then choose four different numbers using simple random sampling. All members of the four departments with those numbers are the cluster sample.



Systematic Sample

- To choose a systematic sample, randomly select a starting point and take every n th piece of data from a listing of the population.
- For example, suppose you have to do a phone survey. Your phone book contains 20,000 residence listings. You must choose 400 names for the sample. Number the population 1–20,000 and then use a simple random sample to pick a number that represents the first name in the sample. Then choose every fiftieth name thereafter until you have a total of 400 names (you might have to go back to the beginning of your phone list). Systematic sampling is frequently chosen because it is a simple method.



Convenience Sampling

- A type of sampling that is non-random is convenience sampling. Convenience sampling involves using results that are readily available.
- For example, a computer software store conducts a marketing study by interviewing potential customers who happen to be in the store browsing through the available software. The results of convenience sampling may be very good in some cases and highly biased (favor certain outcomes) in others.

With Replacement Or Without Replacement

- True random sampling is done with replacement. That is, once a member is picked, that member goes back into the population and thus may be chosen more than once.
- However for practical reasons, in most populations, simple random sampling is done without replacement. Surveys are typically done without replacement. That is, a member of the population may be chosen only once.



Sampling Errors and Non-sampling errors

- When you analyze data, it is important to be aware of sampling errors and non-sampling errors. The actual process of sampling causes sampling errors.
- For example, the sample may not be large enough. Factors not related to the sampling process cause non-sampling errors. A defective counting device can cause a non-sampling error.
- In reality, a sample will never be exactly representative of the population so there will always be some sampling error. As a rule, the larger the sample, the smaller the sampling error.

Critical Evaluation

- We need to evaluate the statistical studies we read about critically and analyze them before accepting the results of the studies. Common problems to be aware of include:
 - Problems with samples: A sample must be representative of the population. A sample that is not representative of the population is biased. Biased samples that are not representative of the population give results that are inaccurate and not valid.
 - Self-selected samples: Responses only by people who choose to respond, such as call-in surveys, are often unreliable.
 - Sample size issues: Samples that are too small may be unreliable. Larger samples are better, if possible. In some situations, having small samples is unavoidable and can still be used to draw conclusions. Examples: crash testing cars or medical testing for rare conditions
 - Undue influence: collecting data or asking questions in a way that influences the response
 - Non-response or refusal of subject to participate: The collected responses may no longer be representative of the population. Often, people with strong positive or negative opinions may answer surveys, which can affect the results.
 - Causality: A relationship between two variables does not mean that one causes the other to occur. They may be related (correlated) because of their relationship through a different variable.
 - Self-funded or self-interest studies: A study performed by a person or organization in order to support their claim. Is the study impartial? Read the study carefully to evaluate the work. Do not automatically assume that the study is good, but do not automatically assume the study is bad either. Evaluate it on its merits and the work done.
 - Misleading use of data: improperly displayed graphs, incomplete data, or lack of context
 - Confounding: When the effects of multiple factors on a response cannot be separated. Confounding makes it difficult or impossible to draw valid conclusions about the effect of each factor.



Example

- A study is done to determine the average tuition that San Jose State undergraduate students pay per semester. Each student in the following samples is asked how much tuition he or she paid for the Fall semester. What is the type of sampling in each case?
 - a. A sample of 100 undergraduate San Jose State students is taken by organizing the students' names by classification (freshman, sophomore, junior, or senior), and then selecting 25 students from each.
 - b. A random number generator is used to select a student from the alphabetical listing of all undergraduate students in the Fall semester. Starting with that student, every 50th student is chosen until 75 students are included in the sample.
 - c. A completely random method is used to select 75 students. Each undergraduate student in the fall semester has the same probability of being chosen at any stage of the sampling process.
 - d. The freshman, sophomore, junior, and senior years are numbered one, two, three, and four, respectively. A random number generator is used to pick two of those years. All students in those two years are in the sample.
 - e. An administrative assistant is asked to stand in front of the library one Wednesday and to ask the first 100 undergraduate students he encounters what they paid for tuition the Fall semester. Those 100 students are the sample.



Variation

- Variation is present in any set of data.
- For example, 16-ounce cans of beverage may contain more or less than 16 ounces of liquid. In one study, six 16 ounce cans were measured and produced the following amount (in ounces) of beverage:
 - 15.8 16.1 15.2 14.8 15.8 15.9
- Measurements of the amount of beverage in a 16-ounce can may vary because different people make the measurements or because the exact amount, 16 ounces of liquid, was not put into the cans. Manufacturers regularly run tests to determine if the amount of beverage in a 16-ounce can falls within the desired range.



Levels of Measurement

- The way a set of data is measured is called its level of measurement.
- Nominal scale level
- Ordinal scale level
- Interval scale level
- Ratio scale level



Nominal scale

- Data that is measured using a nominal scale is qualitative (categorical). Categories, colors, names, labels and favorite foods along with yes or no responses are examples of nominal level data. Nominal scale data are not ordered.
- For example, trying to order people according to their favorite food does not make any sense. Putting veggie pizza first and vegan sushi second is not meaningful.

Ordinal Scale

- Data that is measured using an ordinal scale is similar to nominal scale data but there is a big difference. The ordinal scale data can be ordered.
- An example of ordinal scale data is a list of the top five national parks in the United States. The top five national parks in the United States can be ranked from one to five but we cannot measure differences between the data.

Interval Scale

- Data that is measured using the interval scale is similar to ordinal level data because it has a definite ordering but there is a difference between data.
- Temperature scales like Celsius (C) and Fahrenheit (F) are measured by using the interval scale.

Ratio Scale

- Data that is measured using the ratio scale takes care of the ratio problem and gives you the most information. Ratio scale data is like interval scale data, but it has a 0 point and ratios can be calculated.
- For example, four multiple choice statistics final exam scores are 80, 68, 20 and 92 (out of a possible 100 points). The exams are machine-graded. The data can be put in order from lowest to highest: 20, 68, 80, 92.



Level of Measurement	Put data in categories	Arrange data in order	Subtract data values	Determine if one data value is a multiple of another
Nominal	Yes	No	No	No
Ordinal	Yes	Yes	No	No
Interval	Yes	Yes	Yes	No
Ratio	Yes	Yes	Yes	Yes

Frequency and Frequency Table

- Twenty students were asked how many hours they worked per day. Their responses, in hours, are as follows:
- 5 6 3 3 2 4 7 5 2 3 5 6 5 4 4 3 5 2 5 3.
- Table lists the different data values in ascending order and their frequencies.

DATA VALUE	FREQUENCY
2	3
3	5
4	3
5	6
6	2
7	1

Frequency and Frequency Table

- A relative frequency is the ratio (fraction or proportion) of the number of times a value of the data occurs in the set of all outcomes to the total number of outcomes. To find the relative frequencies, divide each frequency by the total number of students in the sample—in this case, 20. Relative frequencies can be written as fractions, percents, or decimals.

Frequency and Frequency Table

- Cumulative relative frequency is the accumulation of the previous relative frequencies.

The FREQ Procedure

y	Frequency	Percent	Cumulative Frequency	Cumulative Percent
A	3	37.50	3	37.50
B	2	25.00	5	62.50
C	3	37.50	8	100.00

Experiment

- Does aspirin reduce the risk of heart attacks? Is one brand of fertilizer more effective at growing roses than another? Is fatigue as dangerous to a driver as the influence of alcohol? Questions like these are answered using randomized experiments. We will look at important aspects of experimental design. Proper study design ensures the production of reliable, accurate data.

Experiment

- The purpose of an experiment is to investigate the relationship between two variables. When one variable causes change in another, we call the first variable the **explanatory variable**. The affected variable is called the **response variable**. In a randomized experiment, the researcher manipulates values of the explanatory variable and measures the resulting changes in the response variable. The different values of the explanatory variable are called **treatments**. An **experimental unit** is a single object or individual to be measured.

Experiment

- Researchers want to investigate whether taking aspirin regularly reduces the risk of heart attack. Four hundred men between the ages of 50 and 84 are recruited as participants. The men are divided randomly into two groups: one group will take aspirin, and the other group will take a placebo. Each man takes one pill each day for three years, but he does not know whether he is taking aspirin or the placebo. At the end of the study, researchers count the number of men in each group who have had heart attacks.
- Identify the following values for this study: population, sample, experimental units, explanatory variable, response variable, treatments.

Experiment

- The population is men aged 50 to 84.
- The sample is the 400 men who participated.
- The experimental units are the individual men in the study.
- The explanatory variable is oral medication.
- The treatments are aspirin and a placebo.
- The response variable is whether a subject had a heart attack.

Ethics

- The U.S. Department of Health and Human Services oversees federal regulations of research studies with the aim of protecting participants. When a university or other research institution engages in research, it must ensure the safety of all human subjects. For this reason, research institutions establish oversight committees known as Institutional Review Boards (IRB). All planned studies must be approved in advance by the IRB. Key protections that are mandated by law include the following:

Ethics

- Risks to participants must be minimized and reasonable with respect to projected benefits.
- Participants must give informed consent. This means that the risks of participation must be clearly explained to the subjects of the study. Subjects must consent in writing, and researchers are required to keep documentation of their consent.
- Data collected from individuals must be guarded carefully to protect their privacy.

Questions?